

Class Teacher _____ Name _____

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS

Year 11 Preliminary HSC Course

ASSESSMENT TASK 3

September 2013

Time Allowed: 90 minutes

Instructions:

- Write using blue or black pen.
 - Approved calculators may be used.
 - Attempt all questions.
 - All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
 - Marks indicated are a guide only and may be varied if necessary.
- Start each question on a new side of a page.

Total Marks 73

Section 1 Multiple Choice 5 Marks Answer on sheet Allow 8 minutes for this section	Section 2 68 Marks Allow 82 minutes for this section
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1 What is the acute angle between the lines $2x - y - 7 = 0$ and $3x - 5y - 2 = 0$?

- (A) $4^\circ 24'$
- (B) $32^\circ 28'$
- (C) $57^\circ 32'$
- (D) $85^\circ 36'$

2 If $t = \tan \frac{\theta}{2}$ which of the following expressions is equivalent to $4 \sin \theta + 3 \cos \theta + 5$?

- (A) $\frac{2(t+2)^2}{1-t^2}$
- (B) $\frac{(t+4)^2}{1-t^2}$
- (C) $\frac{2(t+2)^2}{1+t^2}$
- (D) $\frac{(t+4)^2}{1+t^2}$

3 The interval DE , where D is $(4, 5)$ and E is $(19, -5)$, is divided internally in the ratio $2:3$ by the point (x, y) . What are the values of x and y ?

- (A) $(-16, 25)$
- (B) $(10, 1)$
- (C) $(12, 0)$
- (D) $(13, -1)$

4 What is the solution to the inequality? $\frac{3}{x-2} \leq 4$

- (A) $x < -2$ and $x \geq -\frac{11}{4}$
- (B) $x > -2$ or $x \leq -\frac{11}{4}$
- (C) $x < 2$ or $x \geq \frac{11}{4}$
- (D) $x > 2$ and $x \leq \frac{11}{4}$

5 Consider the polynomial $P(x) = 3x^3 + 3x + a$.

If $x - 2$ is a factor of $P(x)$, what is the value of a ?

- (A) -30
- (B) -18
- (C) 18
- (D) 30

Question 6 (11 Marks)

Start a new page

Marks

- a) A tangent to a curve makes an angle of 60° to the positive x axis. What is the exact gradient of this tangent. 1
- b) Write $\sec \theta$ in terms of t where $t = \tan \frac{\theta}{2}$ 1
- c) Find a and b if $\frac{\sqrt{7} - 5}{-2 + 3\sqrt{7}} = a + b\sqrt{7}$ 2
- d) By finding an expression for the tangent of the angle between pairs of lines, prove that the triangle formed by the intersection of these 3 lines is an isosceles triangle. You must make your conclusions clear. 3

$$x - \sqrt{3}y + 5 = 0 \text{ and}$$

$$x - y + 10 = 0 \text{ and}$$

$$\sqrt{3}x - y + 10 = 0$$

- e) Consider the curve with equation $y = \frac{x-1}{x+2}$.
- i. Determine the equation(s) of any asymptotes 2
- ii State the range of the function. 1
- iii State the domain of the function. 1

End of Question 6

Question 7 (11 Marks)

Start a new page

- a) If $2x^2 - 9x + 9 \equiv (ax - b)(x - b)$ for all values of x ,
find the values of a and b 2
- b) Derive the equation $\sin 2\theta$ and use this to find the exact value
of $\cos 15^\circ \sin 15^\circ$ 2
- c) Find the cartesian equation of the curve:
 $x = \sin t$ and $y = \sec t$ 2
- d) Find the equation of the tangent line to the curve $y = x^4 + 3x^2 - 1$ at $x = 1$. 2
- e) Find the locus of a point $P(x, y)$ which is equidistant from $3x + 4y = 36$
and $4x + 3y = 24$ 3

End of Question 7

Question 8 (11 Marks)

Start a new page

Marks

- a) Using the auxillary angle method, or otherwise, solve
 $6\sin\theta + 8\cos\theta = 4$ for $0^\circ \leq \theta \leq 360^\circ$

3

- b) Differentiate and write in simplest factorised form:

i) $(3x^2 + 2)^{10}$

2

ii) $\frac{2x^2}{\sqrt{(x^2 - 4)}}$

2

- c) By fully factorizing $f(x)$ (where possible), sketch the graph of
 $f(x) = x^3 - 4x^2 + 8x - 8$ *identifying all key points*

4**End of Question 8**

- a) Solve for x : $3^{2x} = 6(3)^{x-1} + 3$ 2
- b) If α, β, γ are the roots of $3x^3 - 4x^2 + 7x - 11 = 0$
Find:
- i) $\alpha + \beta + \gamma$ 1
- ii) $\alpha\beta\gamma$ 1
- iii) $(\alpha+1)(\beta+1)(\gamma+1)$ 2
- c) By equating the coefficients of $\sin x$ and $\cos x$, or otherwise, find constants A,B satisfying the identity:
 $A(2\sin x + \cos x) + B(2\cos x - \sin x) \equiv \sin x + 8\cos x$ 2
- d) Prove the trigonometric identity: $\frac{\cos 2x}{(\cos x + \sin x)^3} = \frac{\cos x - \sin x}{1 + \sin 2x}$. 2

End of Question 9

Question 10 (11 Marks)

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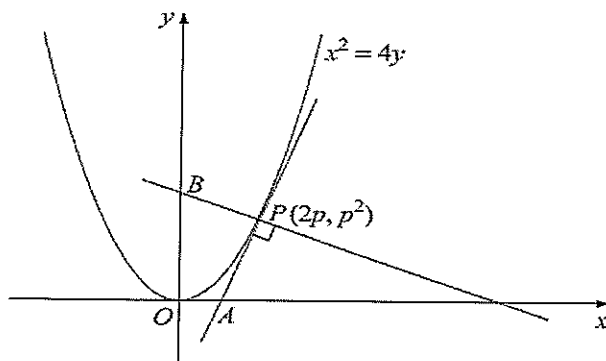
- a) Let $P(x) = (x + 1)(x - 3)Q(x) + ax + b$ where $Q(x)$ is a polynomial and a and b are real numbers. The polynomial $P(x)$ has a factor of $(x - 3)$. When $P(x)$ is divided by $x + 1$ the remainder is 8.

i. Find the values of a and b 2

ii. Find the remainder when $P(x)$ is divided by $(x^2 - 2x - 3)$ 1

- b) From what external point are the tangents to the parabola $x^2 = 8y$ to be drawn so that $4y = 2x + 4$ is the equation of the chord of contact? 2

c)



The diagram shows the graph of the parabola $x^2 = 4y$. The tangent to the parabola at $P(2p, p^2)$, $p > 0$, cuts the x axis at A . The normal to the parabola at P cuts the y axis at B .

- i) Show that the equation of the tangent AP is $y = px - p^2$ 2
(Show all working)

ii) Find the coordinates of A 1

iii) Show that B has coordinates $(0, p^2 + 2)$ 1

- iv) Let C be the midpoint of AB .
Find the Cartesian equation of the locus of C 2

End of Question 10

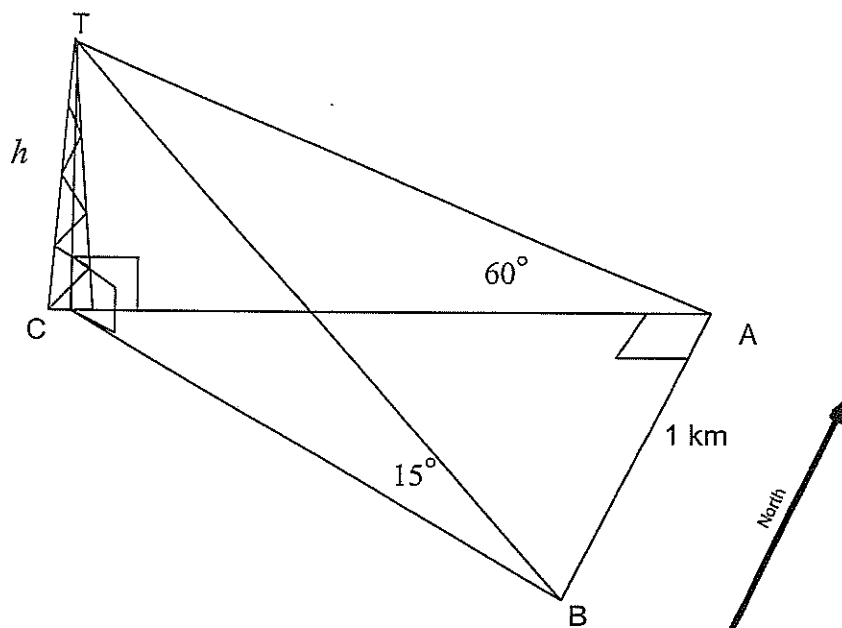
Question 11 (12 Marks)

Start a new page

- a) Find the equation of the normal to the parabola $x^2 = 12y$ at the point where $x = -2$

2

- b) The angle of elevation of the top of a tower (T) from a point A due East of the tower is 60° . From a point B due South of A, the angle of elevation of T is 15° . A and B are at the same elevation as the base of the tower.



The distance $AB = 1$ km

- i) Show that $3h^2 \cot^2 15^\circ = 3 + h^2$
- ii) Find the value of h to the nearest metre
- c) Show algebraically that the line $y = x - 4$ is a tangent to the circle $x^2 + y^2 = 8$ and find the coordinates of the point of contact

2

2

3

Question 11 continues over the page

- d) A piece of wire 24 metres long is cut into two parts, one of which is used to form a square, and the other to form a rectangle whose length is three times its width.
- i) If the width of the rectangle is x show that A , the sum of the areas of the rectangle and the square is given by $A = 7x^2 - 24x + 36$ 1
- ii) Find the vertex of the parabola $A = 7x^2 - 24x + 36$ 1
- iii) Find the lengths of the two parts if the sum of the areas given in part (i) is a minimum.
Do not use calculus 1

End of Question 11

$$\tan 60 = \sqrt{3}$$

$$\sec \theta = \frac{1}{\cos \theta} \quad \cos \theta = \frac{1-b^2}{1+b^2}$$

$$\frac{\sqrt{1-b^2}}{3\sqrt{1-b^2}} \times \frac{3\sqrt{1-b^2}+2}{3\sqrt{1-b^2}} = \frac{2+2\sqrt{1-b^2}-1-b^2}{6-4}$$

$$= \frac{1-1+b^2}{2} = \frac{b^2}{2}$$

$$a = \frac{1}{\sqrt{5}} \quad b = -\frac{3}{\sqrt{5}}$$

$$x - \sqrt{3}y + 5 = 0 \quad \text{--- (1)}$$

$$+\sqrt{3}y = -x + 5$$

$$y = \frac{-x+5}{\sqrt{3}}$$

$$m_1 = \frac{-1}{\sqrt{3}}$$

$$x - y + 10 = 0$$

$$y = x + 10$$

$$m_2 = 1$$

$$\tan \theta = \frac{m_2 - m_1}{1 + (m_1)(m_2)}$$

$$= \frac{1 - \frac{-1}{\sqrt{3}}}{1 + (\frac{-1}{\sqrt{3}})(1)}$$

$$= \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}}$$

$$= \left| \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} \right|$$

$$\theta = 15^\circ$$

$$(ii) \sqrt{3}x - y + 10 = 0$$

$$y = \sqrt{3}x + 10$$

$$m_1 = \sqrt{3}$$

$$y = x + 10$$

$$m_2 = 1$$

$$\tan \theta = \left| \frac{1 - \sqrt{3}}{1 + \sqrt{3}} \right|$$

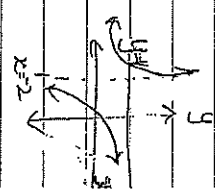
$$\theta = 15^\circ$$

∴ isosceles triangle

$$(c) y = \frac{x-1}{x+2}$$

$$= \frac{x+2-3}{x+2}$$

$$= 1 - \frac{3}{x+2}$$



$$(i) y = 1 \quad VA \quad x = -2$$

$$(ii) R: y < 1 \quad y > 1$$

$$(iii) D: x < -2 \quad x > -2$$

$$(17) 2x^2 - 9x + 9 = (ax-b)(x-b)$$

$$ax^2 - abx - bx + b^2$$

$$ax^2 - x(ab+b) + b^2$$

equating

$$ab+ab = 19$$

$$2ab + b = 19$$

$$3b = 19$$

$$b = \frac{19}{3}$$

$$a = 2$$

$$b = \frac{19}{3}$$

$$(18) \sin(\theta + \phi) = \sin \theta \cos \phi + \cos \theta \sin \phi$$

$$= 2 \sin \theta \cos \theta$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\frac{1}{2} \sin 2\theta = \sin \theta \cos \theta$$

$$\cos 15 \sin 15 = \frac{1}{2} \sin 2\theta$$

$$= \frac{1}{2} \sin 30$$

$$= (\frac{1}{2})(\frac{1}{2})$$

$$= \frac{1}{4}$$

$$(c) x = \sin t \quad y = \sec t$$

$$x^2 = \sin^2 t$$

$$y^2 = 1$$

$$\cos^2 t$$

$$y^2 = \frac{1}{1 - \sin^2 t}$$

$$y^2 = \frac{1}{1 - x^2}$$

$$y = \frac{1}{\pm \sqrt{1-x^2}}$$

$$(d) y = x^4 + 3x^2 - 1$$

$$\frac{dy}{dx} = 4x^3 + 6x$$

$$at x=1 \quad m=10$$

$$y=3$$

$$y-3 = 10(x-1)$$

$$y-3 = 10x-10$$

$$y = 10x-7$$

$$10x-y-7=0$$

$$(e) 3x + 4y = 36 \quad 4x + 3y = 24$$

$$\frac{Ax+By+C}{\sqrt{A^2+B^2}} = \frac{Ax_1+By_1+C}{\sqrt{A^2+B^2}}$$

$$\frac{3x+4y-36}{\sqrt{9+16}} = \frac{4x+3y-24}{\sqrt{16+9}}$$

$$(f) \frac{1}{5}(3x+4y-36) = \frac{1}{5}(4x+3y-24)$$

$$-x-y-12=0$$

$$x-y+12=0$$

(g)

$$-3x-4y+36 = 4x+3y-24$$

$$-7x-7y+60=0$$

$$7x+7y-60=0$$

08

a) $6 \sin \theta + 8 \cos \theta = 4$

$r = 10 \quad \alpha = 53^\circ 8'$

$10 \sin(\theta + 53^\circ 8') = 4$

$\sin(\theta + 53^\circ 8') = 0.4$
 $\theta + 53^\circ 8' = 23^\circ 35'$

$\theta = 103^\circ 17', 330^\circ 8'$

OR $\frac{25}{6(1+t^2)} + 8 \frac{(1-t^2)}{(1+t^2)} = 4$

$12t + 8 - 8t^2 = 4(1+t^2)$

$-12t^2 + 12t + 4 = 0$

$3t^2 - 3t - 1 = 0$

$t = \frac{3 \pm \sqrt{9 - 4(3)(-1)}}{6}$

$= \frac{3 \pm \sqrt{21}}{6}$

$\tan \frac{\theta}{2} = \frac{3 + \sqrt{21}}{6} \quad \tan \theta = \frac{3 - \sqrt{21}}{6}$

$\theta/2 = 51^\circ 39'$

$\theta = 103^\circ 18'$

$\theta = 330^\circ 27'$

b) (i) $y = (3x^2 + 2)^{10}$

$\frac{dy}{dx} = (10)(3x^2 + 2)(6x)$

(ii) $y = (2x^2)(x^2 - 4)^{-1/2}$

$u = 2x^2 \quad v = (x^2 - 4)^{-1/2}$

$u' = 4x \quad v' = (-1/2)(x^2 - 4)^{-3/2}(2x)$

$v' = (-x)(x^2 - 4)^{-3/2}$

$\frac{dy}{dx} = (2x^2)(-x)(x^2 - 4)^{-3/2} + (4x)(x^2 - 4)^{-1/2}$

$= \frac{(2x^3)(-x)}{\sqrt{(x^2 - 4)^3}} + \frac{4x}{\sqrt{x^2 - 4}}$

$= \frac{-2x^4 + 4x(x^2 - 4)}{\sqrt{x^2 - 4}(x^2 - 4)}$

$= \frac{-2x^3 + 4x^3 - 16x}{\sqrt{x^2 - 4}(x^2 - 4)}$

$= \frac{2x^3 - 16x}{\sqrt{x^2 - 4}(x^2 - 4)}$

$(x^2 - 4)(\sqrt{x^2 - 4})$

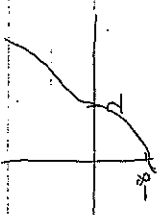
(d) $f(x) = x^3 - 4x^2 + 8x - 8$

$f(2) = 0$

$f(x) = (x - 2)(x^2 - 2x + 4)$

$\Delta = 4 - 4(1)(4)$

$\Delta < 0$ no real roots



09

(a) $3^{2x} = 6(3)^{x-x} \neq 3$

$3^{2x} = 6(3)(3^{-x}) + 3$

$3^{2x} = (6) \frac{3^x}{3} + 3$

$3^{2x} = (2)(3^x) + 3$

Let $u = 3^x$

$u^2 - 2u - 3 = 0$

$(u - 3)(u + 1) = 0$

$u = 3 \quad u = -1$

$3^x = 3$

$x = 1$

no soln.

(b) $3x^3 - 4x^2 + 7x - 11 = 0$

(i) $\alpha + \beta + \gamma = -\frac{b}{a} = \frac{4}{3}$

(ii) $\alpha\beta\gamma = -\frac{d}{a} = \frac{11}{3}$

(iii) $(\alpha + 1)(\beta + 1)(\gamma + 1)$

$= (\alpha\beta + \beta + \alpha + 1)(\gamma + 1)$

$= \frac{11}{3} + \frac{7}{3} + \frac{4}{3} + 1$

$= \frac{22}{3} + 1$

$= \frac{29}{3} + 1 = 8\frac{1}{3}$

(c)

$A(2\sin x + \cos x) + B(2\cos x - \sin x)$

$2A\sin x + A\cos x + 2B\cos x - B\sin x$

$\sin x(2A - B) + \cos x(A + 2B) = \sin x + 8\cos x$

$2A - B = 1 \quad \text{--- (1)}$

$A + 2B = 8 \quad \text{--- (2)}$

$-x2$

$2A - B = 1 \quad \text{--- (1)}$

$(2A + 4B = 16) \quad \text{--- (3)}$

$-5B = -15$

$B = 3 \quad \text{sub into (1)}$

$2A - 3 = 1$

$2A = 4$

$A = 2$

(d) $\frac{\cos 2x}{(\cos x + \sin x)^3} = \frac{\cos x - \sin x}{1 + \sin 2x}$

LHS

$\frac{\cos^2 x - \sin^2 x}{(\cos x + \sin x)^3}$

$\frac{(\cos x - \sin x)(\cos x + \sin x)}{(\cos x + \sin x)^3}$

$\frac{(\cos x - \sin x)}{(\cos x + \sin x)^2}$

$= \frac{\cos x - \sin x}{\cos^2 x + 2\sin x \cos x + \sin^2 x}$

$= \frac{\cos x - \sin x}{1 + 2\sin x} = \text{RHS}$

Q10

(i) $P(x) = (x+1)(x-3)(9x) + ax + b$

(ii) $3a + b = 0 \quad \text{--- (1)}$

$P(1) = (-a + b = 8) \quad \text{--- (2)}$

$a = -2 \quad \text{into (1)}$

$-6 + b = 0$

$b = 6$

$a = -2 \quad b = 6$

(ii) $P(x) = (x+1)(x-3)(9x) + 2x + 6$
Remainder $-2x + 6$

(i) $x^2 = 8y \quad a = 2$

$4y = 2x + 4 \quad x \text{ or } y = 2x(y+4)$
 $2x = 4y - 4$
 $= (2)(2)(y-1)$

Point $(2, -1)$

(i) $x^2 = 4y \quad P(2p, p^2)$

$x^2 = 4y \quad \text{Eqn tangent :-}$
 $y - p^2 = p(x - 2p)$
 $y - p^2 = px - 2p^2$
 $mx = p \quad y = px - p^2$

(ii) at A $y = 0$

$0 = px - p^2$
 $px = p^2$
 $x = p$

$(p, 0)$

(iii) Eqn normal $m_1 = -1/p$

$y - p^2 = -1/p(x - 2p)$
 $yp - p^3 = -x + 2p$

at $x = 0 \quad yp = p^3 + 2p$
 $y = p^2 + 2$

$B(0, p^2 + 2)$

(iii) co-ords A

$y = px - p^2 \quad \text{at } y = 0$
 $0 = px - p^2$
 $x = p \quad (p, 0)$

mid point A $(p, 0)$ B $(0, p^2 + 2)$

$x = \frac{0+p}{2} \quad y = \frac{p^2+2}{2}$
 $2x = p \quad y = \frac{(2x)^2 + 2}{2}$
 $y = \frac{4x^2 + 2}{2}$
 $y = 2x^2 + 1$

Q11

(i) $x^2 = 12y$

$y = x^2/12$

$y' = 2x/12 \quad \text{at } x = -2 \quad y = 1/3$
 $m = -1/3 \quad m_{\text{normal}} = 3$

$y - 1/3 = 3(x + 2)$
 $3y - 1 = 9x + 18$
 $9x - 3y + 19 = 0$

(b) $\tan 60 = h/AC \quad \tan 15 = h/BC$

(i) $AC = h/\sqrt{3} \quad BC = h/\tan 15$

$BC^2 = AB^2 + AC^2$
 $h^2 = 1 + \frac{h^2}{3}$
 $\tan^2 15$

$3h^2 \cot^2 15 = 3 + h^2$

(ii) $3h^2 = 3 + h^2$
 $\tan^2 15$

$3h^2 = 3 + h^2 (\tan^2 15)$
 $3h^2 = 3 \tan^2 15 + h^2 \tan^2 15$
 $3h^2 - h^2 \tan^2 15 = 3 \tan^2 15$
 $h^2 (3 - \tan^2 15) = 3 \tan^2 15$
 $h^2 = \frac{3 \tan^2 15}{3 - \tan^2 15}$
 $= \frac{271.2 \text{ m}}{271 \text{ m}}$

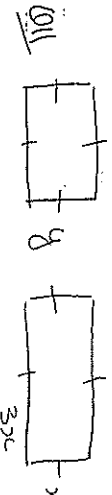
(c) --

$y = x - 4 \quad x^2 + y^2 = 8$

$x^2 + (x - 4)^2 = 8$
 $x^2 + x^2 - 8x + 16 - 8 = 0$
 $x^2 - 4x + 4 = 0$

$\Delta = b^2 - 4ac$
 $= 16 - (4)(1)(4)$
 $= 0 \quad \therefore \text{tangent}$

$x^2 - 4x + 4 = 0$
 $(x - 2)^2 = 0$
 $x = 2 \quad y = -2$
 $(2, -2)$



$4y + 8x = 24$
 $4y = 24 - 8x$
 $y = 6 - 2x$

$A = y^2 + 3x^2$
 $= (6 - 2x)^2 + 3x^2$
 $= 36 - 24x + 4x^2 + 3x^2$
 $= 7x^2 - 24x + 36$

(iii) square :-
 $x = \frac{-b}{2a}$
 $= \frac{24}{2 \times 7}$
 $= \frac{12}{7}$
rectangle :-
 $15/7 \times 5/7$

